Mahindra University Hyderabad
École Centrale School of Engineering
End Semester Examinations,
December-2025

Program: B. Tech. Branch: Computation & Mathematics Year: II Semester: I
Subject: Algebra (MA 2106)

Date: 9/12/2025 **Start Time: 10:00 AM**
Time Duration: 3 Hours **Max. Marks: 100**

Instructions:

1. All questions are compulsory. The marks assigned to each question are indicated at the end of the question.
2. Provide justification for your answers wherever required.
3. All the notations are standard as used in class lectures.

1. Each of the following multiple choice question has four options, out of which only one is correct. Choose the correct option: [2 × 10]

   (i) In a ring R, an ideal I is maximal if:
   - A. If every element of I is a unit
   - B. The quotient R/I is a field
   - C. I is not contained in another subring
   - D. I is a field

   (ii) Which of the following is not a group?
   - A. $\mathbb{Q} \setminus \{0\}$ under multiplication
   - B. $\mathbb{Z}$ under addition
   - C. $\mathbb{R}^+$ under multiplication
   - D. $\mathbb{Z}_n$ under multiplication for any $n \geq 1$

   (iii) Which of the following is true:
   - A. A commutative ring, with unity, is a field
   - B. A ring, in which every nonzero element is invertible, is a field
   - C. An integral domain is a field
   - D. None of the above is true

   (iv) The order of the element $(469213)(315824)$ in S_9 is

A. 6
B. 36
C. 3
D. 18

(v) The number of elements of order 3 in S_3 is:
A. 1
B. 2
C. 3
D. 4

(vi) Which of the following statement is not true?
A. Every finite integral domain is a field
B. The characteristic of a field is the additive order of the unity element
C. The kernal of a group homomorphism is always a normal subgroup
D. The image of a group homomorphism is always a normal subgroup

(vii) Which of the following is always true for a ring R?
A. $ab = 0$ implies $a = 0$ or $b = 0$ for all $a, b \in R$
B. Every ring has characteristic 0
C. Every ideal of R contains the unity element
D. Every field is an integral domain

(viii) The order of an element $g \in G$ is:
A. The smallest positive integer n such that $g^n = g$
B. The smallest positive integer n such that $g^n = e$
C. The number of generators in the cyclic subgroup generated by g
D. always a composite number

(ix) In which of the following group does every element commute with every other element?
A. S_4
B. D_{10}
C. $\mathbb{Z}[x]$ under addition
D. $GL(2, \mathbb{R})$

(x) The number of generators in a cyclic group of order 10 is:
A. 1
B. 2
C. 4
D. 8

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2. Find the following: [3 × 10]

(i) The order of the field $\frac{\mathbb{Z}_3[x]}{\langle x^3+2x+1\rangle}$.

(ii) The number of elements in the ring $\frac{\mathbb{Z}_7[i]}{\langle 2i+1\rangle}$.

(iii) The number of elements of order 4 in the dihedral group of order 8.

(iv) The number of elements which have multiplicative inverses in $\mathbb{Z}_{30}$.

(v) The kernel of the homomorphism $f : \mathbb{Z}_{18} \mapsto \mathbb{Z}_{24}$ defined by $f(\overline{t}) = \overline{3t}$.

(vi) The multiplicative inverse of $\overline{5}$ in $\mathbb{Z}_{11}$.

(vii) A generator of the cyclic group $U(18)$.

(viii) The quotient and remainder upon dividing $x^4 + x^3 + 3x + 2$ is divided by $x^2 + x + 1$ in $\mathbb{Z}_5[x]$.

(ix) The number of elements in the ring $\frac{\mathbb{Z}[i]}{\langle 3-2i \rangle}$.

(x) The remainder upon dividing x^{27} by $x + 4$ in $\mathbb{Z}_7[x]$.

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3. (i) Let G be an abelian group. Show that the set $\{a \in G| \ a^3 = e\}$ is a subgroup of G. [2]

(ii) Let $f : G \to G'$ be a group homomorphism. Show that $\ker f = \{e\}$ iff f is injective. [4]

(iii) Let $f : G \to G'$ be a surjective homomorphism and let N be a normal subgroup of G. Show that $f(N)$ is a normal subgroup of G'. [4]

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4. (i) Construct a field of order 125. [05]

(ii) Show that $x^2 + 3x + 3$ is irreducible over $\mathbb{Z}_5$. [03]

(iii) Show that $3x^4 + 7x^3 + 14x^2 + 105$ is irreducible over $\mathbb{Q}$. [05]

(iv) Show that $x^4 + 3x + 1$ is irreducible over $\mathbb{Q}$. [7]

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5. Let

$$R = \left\{ \begin{pmatrix} a & a-b \\ a-b & b \end{pmatrix} | a, b \in \mathbb{Z} \right\}.$$

(i) Show that R is a subring of $M_2(\mathbb{Z})$, where $M_2(\mathbb{Z})$ is the ring of 2×2 matrices with entries from $\mathbb{Z}$. [5]

(ii) Show that R is not an ideal of $M_2(\mathbb{Z})$. [5]

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6. (i) Show that in the ring $\frac{\mathbb{Z}[x]}{\langle 3x+1 \rangle}$, the element $3+ < 3x + 1 >$ is a unit. [3]

(ii) How many elements does the field $\frac{\mathbb{Z}_3[x]}{\langle x^2+x+2 \rangle}$ have? Show that the multiplicative group of the field $\frac{\mathbb{Z}_3[x]}{\langle x^2+x+2 \rangle}$ is cyclic by showing that one of these elements generates all the elements. [7]
